AFDB · DATA INNOVATION FOR AFRICA (DI4A) · NON-TRADITIONAL DATA

Measuring the Connectivity Gap with Ookla Open Data



From crowdsourced speed tests to a population-weighted subnational indicator, with WorldPop

Data Innovation For Africa (DI4A) - AfDB

SECTION 01

01

Why connectivity is a statistical question

Three-quarters of humanity is online. So why is the gap still growing?



Four numbers that frame the problem

6.0 bn

people online in 2025

About 74% of the world's population, up from 5.8 bn in 2024.

2.2 bn

people still offline

96% of them live in low- and middle-income countries.

84% vs 4%

5G population coverage

High-income vs low-income countries (world: 55%).

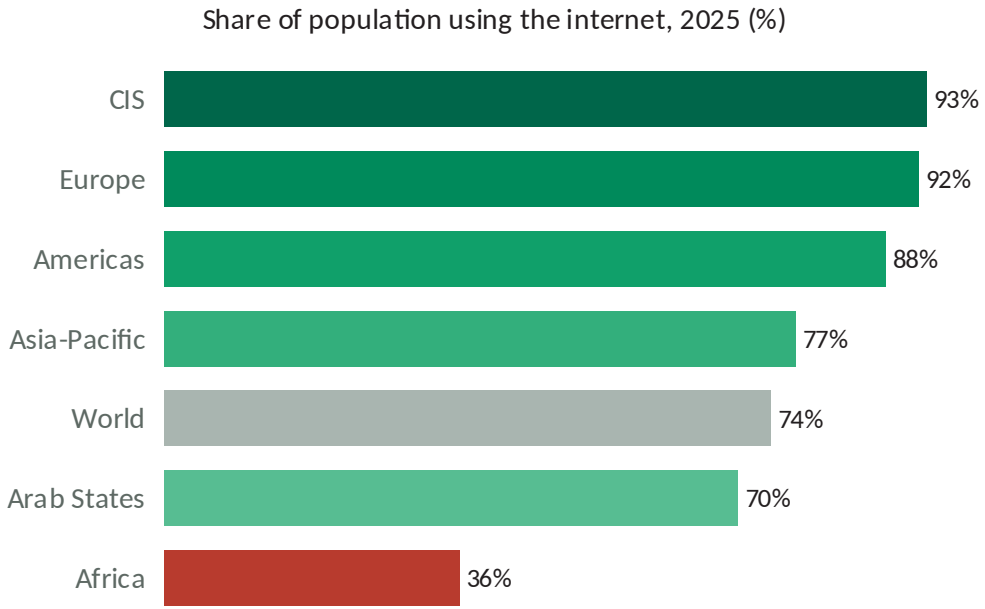
≈ 8×

more mobile data per user

Generated by a typical high-income vs low-income user.

Source: ITU, Measuring digital development: Facts and Figures 2025 (17 November 2025).

Africa is the region with the most to measure



36%

of Africans online

Less than half the global level (74%). Low-income countries: 23%.

2.6×

urban–rural ratio in Africa

Urban residents are 2.6 times more likely to be online than rural ones (Europe: 1.1).

14%

of rural dwellers online

In low-income countries: about one in seven.

Source: ITU, Facts and Figures 2025 (Internet use; Internet use in urban and rural areas). ITU regional groupings.

From “is there a connection?” to “is it usable?”

THE OLD QUESTION

Is there a connection?

- Access treated as binary: served / unserved
- Measured with coverage maps and subscription counts
- SDG 9.c.1 coverage · 17.6.1 fixed broadband subscriptions · 17.8.1 internet use

THE NEW QUESTION

Is the connection usable?

- Access treated as a gradient, not a switch
- Speed, latency, reliability, price and skills
- The ITU target: universal and meaningful connectivity

Where Ookla fits: it measures the experience of real users, not availability. It complements surveys and regulator data; it does not replace them.

Framing after ITU, Facts and Figures 2025. SDG indicators: UN Global SDG Indicator Framework.

SECTION 02

02

Inside the dataset

What exactly does one row of this data represent?



How a speed test becomes an open data tile

Test

Users run Speedtest on Android and iOS apps. Hundreds of millions of tests per month.

Filter

Only results with GPS-quality location accuracy are kept.

Aggregate

Tests are averaged inside zoom-16 tiles (≈ 611 m) to protect testers.

Publish

Quarterly, on AWS, in Shapefile and Parquet, free of charge.

Global
One schema for every country

Granular
≈ 611 m tiles, not provinces

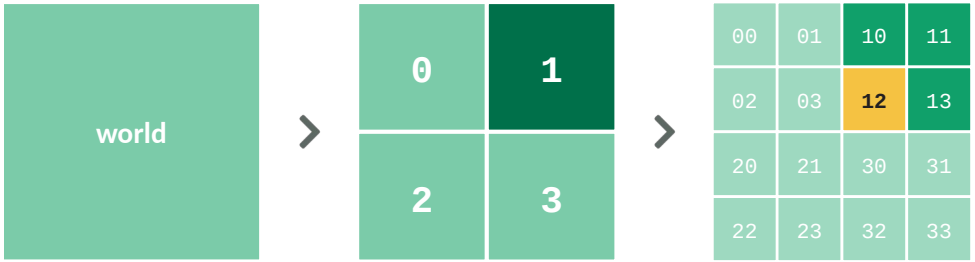
Open
CC BY-NC-SA 4.0

Regular
Quarterly since Q1 2019

Ookla's stated purpose: *make the data easier to use in closing the gaps between people who have modern internet access and those who do not.*

Source: teamookla/ookla-open-data (GitHub README); ooklaOpenDataR package documentation.

The tile and its quadkey, explained



Each level splits a tile into four and adds one digit: 0 ↖ 1 ↗ 2 ↙ 3 ↘

At zoom 16 (the Ookla grid)

`quadkey = 16 digits, e.g. 1220303103213012`

≈ 610.8 m × 610.8 m at the equator (18 arc-second blocks)

Stable ID

Join the same tile across quarters on quadkey, with no spatial operation.

Prefix = parent

All tiles inside a larger tile share its prefix, so country filters are fast.

WGS 84 geometry

Stored in EPSG:4326; the grid itself follows Web Mercator.

Size shrinks

Tiles get smaller in metres away from the equator.

Source: teamookla/ookla-open-data README; Bing Maps Tile System. The 16-digit quadkey shown is illustrative.

What each tile tells you, and the traps

FIELD	MEANING	UNIT / NOTE
`avg_d_kbps`	Average download speed	*kbps ÷ 1000 → Mbps*
`avg_u_kbps`	Average upload speed	*kbps*
`avg_lat_ms`	Average idle latency	*milliseconds*
`avg_lat_down_ms / _up_ms`	Latency under load	*Parquet only · sparse*
`tests`	Number of tests in the tile	*confidence weight*
`devices`	Number of unique devices	*guards against one heavy tester*
`quadkey`	Tile identifier	*join key across quarters*
`tile · tile_x, tile_y`	WKT geometry · centroid	*centroid from Q3 2023*

Unit trap

Speeds are in kbps. 25 000 kbps = 25 Mbps.

Statistic trap

Tile values are means. The Speedtest Global Index reports medians: never mix them.

Sample trap

Always read tests and devices before the speed. Set and state a minimum.

Source: teamookla/ookla-open-data README (Attributes). Loaded-latency fields added 23 Feb 2023; centroids 1 Jul 2023.

Two layers, distributed as separate files

Fixed broadband

`performance_fixed_tiles`

- Tests over Wi-Fi or Ethernet
- Proxy for home and office broadband
- Caution: on Wi-Fi you partly measure the local router, not the access line

Mobile (cellular)

`performance_mobile_tiles`

- Tests over 4G LTE, 5G NR and other cellular technologies
- Proxy for experience where people actually are
- Sensitive to indoor reception, cell load and time of day

File naming: files are named by the first day of the quarter, e.g. `2025-10-01_performance_mobile_tiles.parquet` = Q4 2025.

Source: teamookla/ookla-open-data README.

Seven years of quarters, and what changed

Q1 2019	Feb 2023	Jul 2023	Apr 2026	Today
First quarter published; continuous series since	Latency-under-load fields added (Parquet only)	Tile centroid coordinates tile_x, tile_y added	Supported regions updated: some areas removed	Check the bucket for the latest published quarter

Step zero: verify national coverage

Count tiles and tests for your country in the latest quarter and in the same quarter one year earlier, before designing any indicator.

Re-aggregation for privacy

Data is periodically re-aggregated to honour GDPR, CCPA and LGPD requests. The same quarter downloaded twice can differ: record your access date.

Source: teamookla/ookla-open-data README (changelog, entry of 16 April 2026). Latest available quarter to be confirmed on the day.

SECTION 03

03

Access and licence

Free to download. But free to publish as official statistics?



Four ways to get the files

AWS S3 bucket

s3://ookla-open-data, by format, type, year, quarter.

Direct HTTPS

Any object from a browser. No account needed.

AWS CLI

Scriptable and resumable; no credentials needed.

R package

ooklaOpenDataR, for teams working in R.

DOWNLOAD ONE QUARTER (AWS CLI)

```
export FORMAT=parquet  # or shapefiles
export TYPE=mobile     # or fixed
export YYYY=2025 Q=4

aws s3 cp \
 s3://ookla-open-data/${FORMAT}/performance/\
type=${TYPE}/year=${YYYY}/quarter=${Q}/ . \
 --recursive --no-sign-request
```

Formats: Shapefile for GIS desktops · Parquet (+ WKT) for pandas, DuckDB, Athena

Source: teamookla/ookla-open-data README; teamookla/ooklaOpenDataR.

What the licences mean for an NSO

OOKLA OPEN DATA · CC BY-NC-SA 4.0

BY — Attribution
Credit Ookla, give the data period and your access date.

NC — Non-commercial
No commercial use: paid dashboards, consultancy deliverables.

SA — Share-alike
Distributed derivatives must carry the same licence.

BY — WorldPop · CC BY 4.0
Attribution only; commercial use allowed.

Sources: Ookla Open Data README (licence and attribution); WorldPop FAQ and licence. Not legal advice.

Three questions for your legal unit

1. Can an indicator derived from CC BY-NC-SA data be released under our national dissemination licence?
2. Is a public official publication “non-commercial” if third parties reuse it commercially?
3. Do we need a data-sharing agreement with Ookla or operators for regular production?

SECTION 04

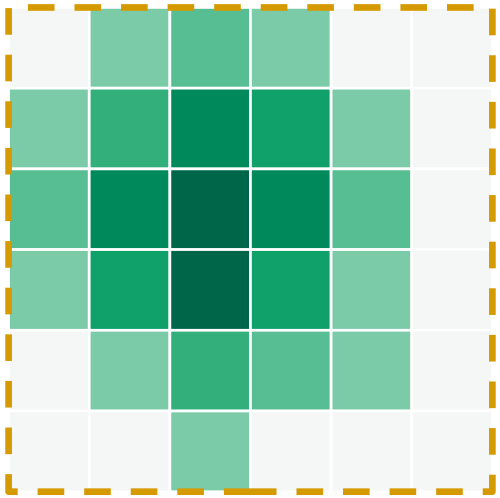
04

Building the indicator

A speed without a population is not yet a statistic.



WorldPop: the denominator Ookla lacks



One Ookla tile (≈ 611 m) holds about 36 WorldPop cells (≈ 100 m). Sum them to get the tile's population.

100 m grid

People per cell at 3 arc-seconds; 1 km versions also exist.

2015 – 2030

Annual estimates, age and sex breakdowns available.

Constrained

Population placed only in cells mapped as built settlement.

UN-aligned

National totals aligned to UN WPP 2024 baselines.

CC BY 4.0

Free to reuse with attribution, GeoTIFF format.

Model output

Not a census; uncertainty rises away from census years.

Source: WorldPop Global2 R2025A v1 release statement and data hub (DOI 10.5258/SOTON/WP00839); WorldPop licence.

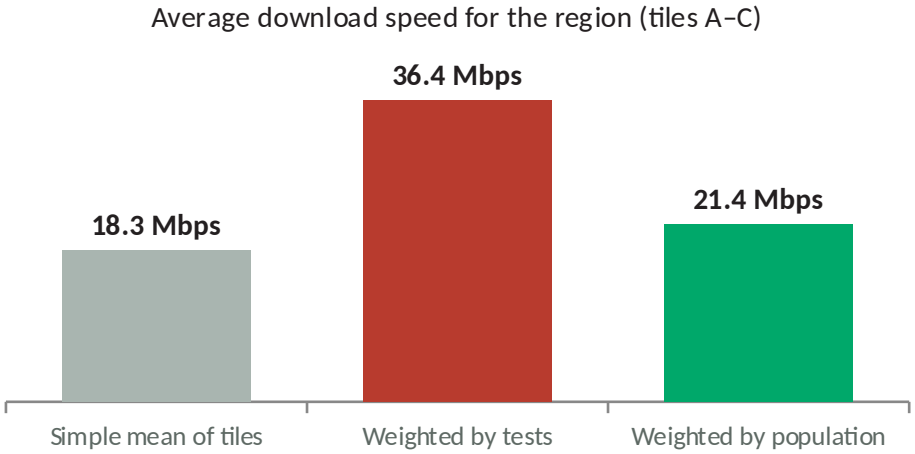
One region, three averages, three stories

TILE	MBPS	TESTS	PEOPLE
A · city	40	120	8 000
B · peri-urban	12	15	6 000
C · rural	3	2	5 000
D · rural	—	0	4 000

Report these two alongside the speed

82.6% of people live in a measured tile (19 000 of 23 000)

60.9% of people live in a tile ≥ 10 Mbps (14 000 of 23 000)



Tests over-weight the city: well-connected people test more.

Population answers the policy question: what does a typical resident get?

Illustrative numbers built for teaching. Population-weighted mean = Σ(speed × people) / Σ people over measured tiles.

Six steps from raw tiles to a regional indicator

1 Load

Read one quarter of the mobile Parquet file.

`pandas`

2 Clip

Keep tiles in the country bounding box, using tile_x / tile_y.

`pandas`

3 Add people

Sum WorldPop cells inside each tile (zonal statistics).

`rasterstats`

4 Assign region

Spatial join of tile centroids to admin boundaries.

`geopandas`

5 Filter

Drop tiles below a stated minimum of tests and devices.

`pandas`

6 Compute & check

Weighted indicators per region, then sanity checks.

`pandas`

Step 0, before all of this: verify national coverage (tiles and tests) for the quarters you plan to use.

The whole pipeline in about twenty lines

```
import pandas as pd, geopandas as gpd
from rasterstats import zonal_stats

URL = ("https://ookla-open-data.s3.amazonaws.com/parquet/"
       "performance/type=mobile/year=2025/quarter=4/"
       "2025-10-01_performance_mobile_tiles.parquet")
adm = gpd.read_file("adm2.gpkg").to_crs(4326)
x0, y0, x1, y1 = adm.total_bounds

df = pd.read_parquet(URL)                          # 1 load
df = df[df.tile_x.between(x0, x1)
        & df.tile_y.between(y0, y1)]               # 2 clip
t = gpd.GeoDataFrame(df, crs=4326,
        geometry=gpd.GeoSeries.from_wkt(df.tile))
t["pop"] = [z["sum"] or 0 for z in zonal_stats(   # 3 people
            t, "worldpop_2025_100m.tif", stats="sum")]
t = gpd.sjoin(t.set_geometry(t.centroid),          # 4 region
              adm[["adm2", "geometry"]])
t = t[(t.tests >= 3) & (t.devices >= 2)]           # 5 filter
t["w"] = t.avg_d_kbps / 1000 * t["pop"]            # 6 compute
pw = t.groupby("adm2")["w"].sum() / t.groupby("adm2")["pop"].sum()
```

Python 3 with pandas, geopandas ≥ 1.0 and rasterstats. File names for boundaries and WorldPop are placeholders.

Filter before geometry

Building WKT geometries for the global file is the slowest step.

Water and empty cells

zonal_stats returns None where no population: treat as 0.

Thresholds are choices

3 tests / 2 devices is an example. Choose, test and document yours.

Full notebook

Available in the DI4A GitHub organisation.

A small, honest set of regional indicators

Population-weighted speed

`Σ (Mbps × pop) / Σ pop`

What a typical resident of measured areas experiences.

Measured-population share

`pop in measured tiles / total pop`

How much of the region the speed actually represents.

Share of people above a threshold

`pop in tiles ≥ X Mbps / total pop`

Easy to communicate. X is a policy choice: state it.

Change between quarters

`indicator(t) − indicator(t−4)`

Year-on-year, same quarter, same rules, joined on quadkey.

Always attach: layer (mobile / fixed) · quarter · access date · minimum tests & devices · WorldPop release and year

SECTION 05

05

Reading the signals

What do the tiles show that a national average hides?



Four patterns to look for in your results

1 Absence

A missing tile is information, not an error. Where do people live with no measurement?

How: Map tile presence over several quarters, against population.

2 Gradient

Performance decays from city centres. Look for the cliff, not the slope.

How: Plot speed by distance band or density decile.

3 Substitution

Tiles with mobile but no fixed tests suggest mobile-dependent households.

How: Join the two layers on quadkey.

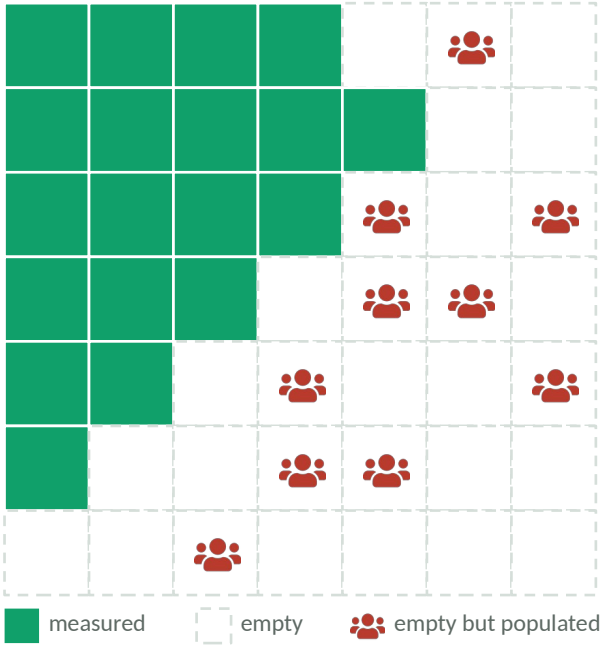
4 Responsiveness

Latency, especially under load, reveals congestion that speed hides.

How: Parquet only, from 2023, sparsely populated.

Latency fields: teamookla/ookla-open-data README.

Absence: the most important value is often missing



Schematic illustration, not real data.

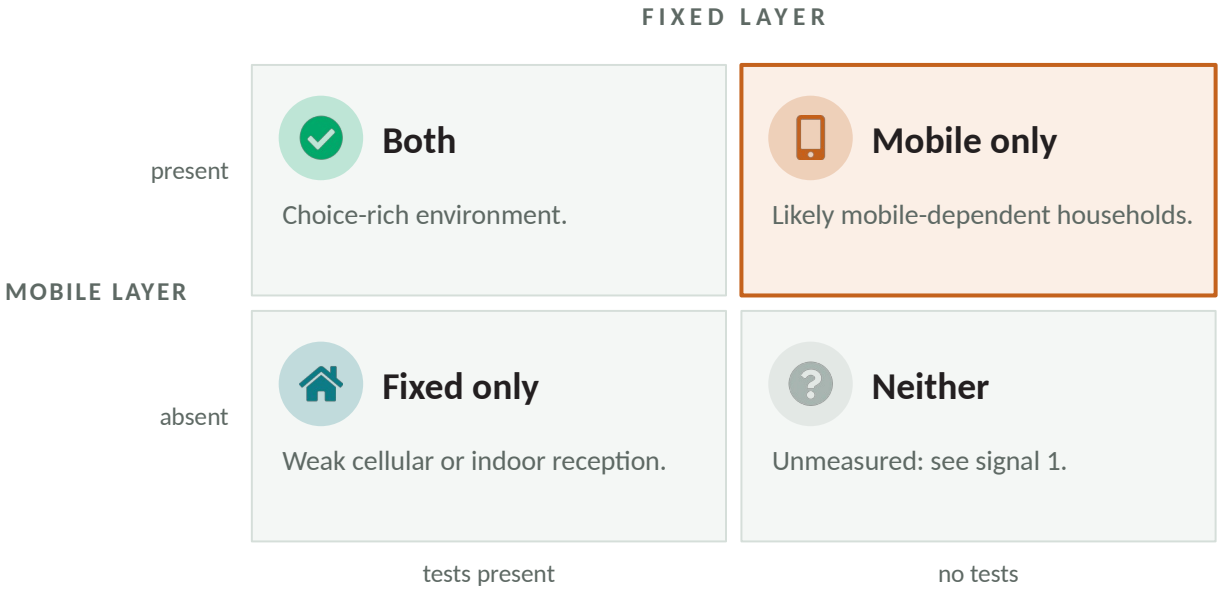
AN EMPTY TILE CAN MEAN

- No service
- Service nobody can afford
- Few or no smartphones
- Almost nobody lives there
- People who never ran a test

How to work with it

- Map tile presence, not only tile values, across several quarters.
- Overlay WorldPop: absence in a populated tile is the case to study.
- Treat persistent absence as a lead for field verification, never as a finding.

Substitution: who lives on mobile alone?



Why it matters

Mobile dependence is invisible in coverage statistics and very visible in household budgets.

This join is one of the few ways open data can surface it, as a hypothesis to test against your household survey.

Join key: quadkey, same quarter. In many African countries the fixed layer is sparse: check counts first.

SECTION 06

06

Rigour, limits and publication

What must be true before a number leaves the office?





What this data can and cannot tell you

It can establish

- Performance actually experienced by real users
- Fine spatial patterns invisible in national statistics
- Change quarter by quarter since 2019
- Fixed vs mobile contrast in the same place
- A floor on where service demonstrably exists

It cannot establish

- Coverage or availability: it is not a coverage map
- Adoption, subscription or penetration rates
- Prices, affordability or household budgets
- Who the users are: no age, sex or income
- The experience of people who never test

Observational, self-selected data: manage the constraint explicitly in every publication.

Who uses this data, and what an NSO could do

DOCUMENTED USERS

Regulators

US FCC and Malaysia's MCMC: operator accountability and connectivity funding.

Education

Georgia (US) Department of Education: placing Wi-Fi school buses.

Research & NGOs

Internet inequality studies; mission connectivity planning.

POSSIBLE NSO USES

Subnational digital-inclusion profiles

Complement survey-based SDG 17.8.1.

Covariate for small-area estimation

Alongside night-time lights data.

Field operations planning

Anticipate CAPI upload constraints by area.

Evidence for data partnerships

Access agreements with Ookla, operators, regulators.

Documented users: Ookla (Ookla for Good). NSO uses: options for discussion, not endorsements.

Sources and further reading

The data

- Ookla: teamookla/ookla-open-data, README, schema, tutorials (GitHub)
- Ookla: ooklaOpenDataR R package (GitHub)
- AWS Registry of Open Data: Speedtest global performance
- WorldPop Global2 R2025A v1, release statement, DOI 10.5258/SOTON/WP00839

Global context

- ITU, Measuring digital development: Facts and Figures 2025 (17 Nov 2025)
- ITU press release, 17 November 2025
- UN Global SDG Indicator Framework: 9.c.1, 17.6.1, 17.8.1

Method

- Bing Maps Tile System (quadkeys)
- Creative Commons licences: CC BY 4.0, CC BY-NC-SA 4.0
- Peer-reviewed literature on sampling bias in crowdsourced speed tests

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Questions, then to the lab

Open your notebook, pick your country, and start with step zero: check the coverage.

